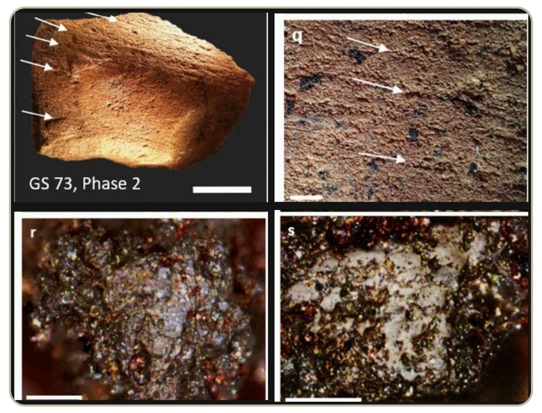
How one stone becomes evidence

Read across for twelve months of growth. **Read down** from the marked phrase, to what it proves, to the continuum descriptor. Only the phrases that prove a criterion are marked — the black text is the glue.



What does GS73 suggest about life at Madjedbebe in ancient Australia?

ANCIENT OBJECT
GS73, a sandstone millstone fragment excavated from square B5/52 at Madjedbebe on Mirarr Country.

MODERN RECORD
Hayes and colleagues published photographs, micrographs and laboratory results in a peer-reviewed study in 2022.

EXCAVATION CONTEXT
Phase 2, modelled to 68.7–50.4 thousand years ago. The earliest end of this range is debated.

MEASURED EVIDENCE
Deep partial grooves; levelled quartz grains; striations; bright polish on the raised parts of grains.

RESIDUES
41 starch grains recovered and plant-related compounds detected. The exact plant was not identified.

KNOWN LIMITS
Two chemical peaks were identified as plastic contamination. The integrity of the Phase 2 layer is contested.

■ **Origin, features & purpose** — marked like this

▲ **Historical context** — marked like this

● **Historical interpretations** — marked like this

◆ **Accuracy, usefulness & reliability** — marked like this

★ **Metacognition** — marked like this

YEAR LEVEL	Grade 5	Year 6	Year 7 EXPECTED	Year 8	Year 9	Year 10
WORKED EXAMPLE	GS73 is a stone source found at Madjedbebe. I can see deep grooves and shiny patches. It comes from ancient Australia and may have been used for grinding. Researchers disagree about exactly how old its soil layer is. The stone is useful for showing tool use, but it cannot tell us the user's name. I checked my answer and added the words 'may have' because the source does not prove everything.	GS73 is a stone tool: a broken piece of a grinding stone from Madjedbebe, on Mirarr Country. Photographs show deep grooves and shiny, polished patches. People were living there many thousands of years ago, and they used stones like this one to grind their food. Some archaeologists believe the layer is about 65,000 years old. Others believe the soil has moved, so the date is not certain. The stone is useful evidence that people were grinding materials, but it cannot tell us which plants they were grinding. I divided the task into four smaller parts and finished one part at a time.	The ancient object is GS73, a broken piece of a sandstone grinding stone excavated at Madjedbebe, on Mirarr Country. The modern source is a set of photographs published by Hayes and her team in 2022. Their purpose was to record the wear and argue that people ground food there for a very long time. In ancient Australia, grinding allowed people to eat hard seeds and tough plants. It also shows they understood the resources around them. Hayes and her team interpret this as plant processing, because the stone carries deep grooves, microscopic polish and 41 starch grains. Other researchers question how old the layer really is. GS73 is very useful for studying how people prepared food, because its shape, its wear and its plant traces all agree, but it cannot tell us which plant was ground. When I reviewed my answer I changed 'proves' to 'supports', and separated the well-supported part from the part that is still argued about.	GS73 is a broken millstone recovered from Phase 2, one of the oldest occupation layers at Madjedbebe. Hayes and her team photographed it and examined it under a microscope. They published the images so that other researchers could inspect the same evidence and judge their interpretation. Madjedbebe is a rock shelter on Mirarr Country that was occupied for many thousands of years. The grinding marks show that people were already preparing plant foods early in Aboriginal history. Hayes explains the wear and the plant residues as signs of seed grinding. Clarkson argues that the artefacts remained where they were dropped, which supports the early date. Williams points to termite activity and moving sediment, which would make that date less reliable. The shape, the microscopic wear, the starch and the plant chemicals all point in the same direction, so the grinding interpretation is well supported. Even so, I would ask whether the residues could be contaminated, and whether the layer could have shifted. Next time I will check which tests support each other independently and record the questions that remain unresolved.	GS73 is material evidence, excavated from a recorded layer called Phase 2. Hayes and her team published photographs, microscope images and laboratory results together. They did this because one form of evidence alone would not convince a scientific audience. The study belongs to wider research into the deep history of Aboriginal Australia, and it was carried out with the permission of Mirarr custodians. Hayes analyses the grooves, the polish and the plant residues as evidence of seed processing. Clarkson analyses the condition of the site and the dating to defend the early chronology. Williams uses signs of termite disturbance and sediment movement to challenge it. Because the wear and the residues agree independently, grinding is the interpretation I would trust most. The exact plant and the precise age are weaker, because both depend on preservation and on how stable the deposit is. While writing, I monitored that I was listing evidence rather than comparing it, and I rewrote my ending as a judgement.	GS73 is an ancient artefact, but we meet it only through a modern scientific publication. Its explicit evidence is the recorded wear and the plant residues preserved on the stone. Its implicit argument is that grinding technology in Australia is far older than earlier accounts allowed, and that argument is aimed at other archaeologists. The investigation reflects modern laboratory testing carried out with Mirarr custodians. It also reveals what those researchers value: testing claims, seeking permission, and stating honestly how certain they are. Hayes and Clarkson accept the early chronology because several independent records converge on it. Williams gives more weight to termite disturbance and sediment movement. Their interpretations differ because they evaluate the formation of the site differently. Overall, GS73 is strong evidence for grinding, but only qualified evidence for a precise age. Its main strength is that separate tests agree; its limits are the unidentified plant, the modern plastic contamination, and the ongoing argument about the layer. I justified relying on the interpretation that several tests supported, then adapted my conclusion so the two claims did not sound equally certain.
WHAT IT PROVES	■ Names the source and two visible features: 'stone source', 'deep grooves' and 'shiny patches'. Origin and purpose are not yet developed. ▲ Places GS73 broadly in 'ancient Australia', but gives little detail about life or practice. ● Notices disagreement about the date without describing the competing arguments. ◆ Gives one use and one limit: tool use is visible, but the user cannot be named. ★ Shows a real revision — adding 'may have' makes the inference appropriately cautious.	■ 'Archaeological artefact', 'sandstone grinding-stone fragment' and 'grooves and polish' identify type and features; purpose is not yet explained. ▲ 'A very early period of Aboriginal occupation' identifies the context and connects it to repeated tool use. ● Recognises two positions on the date, but does not yet describe the evidence behind them. ◆ Recognises usefulness for grinding and a clear limit: the material processed cannot yet be identified. ★ Plans the response by breaking the task into the four historical-thinking parts shown on the wall.	■ Separates the ancient object from the modern 2022 record, then identifies origin, features and why the figure was created. ▲ Outlines how grinding supported plant preparation and knowledge of local resources. ● Describes the plant-processing interpretation and the separate challenge to the earliest date. ◆ Concludes that converging methods make plant processing useful evidence, then bounds the claim with two limits. ★ Makes editing visible: 'proves' becomes 'supports', and function is separated from precise age.	■ Describes the object and explains how the modern figure lets readers inspect its form, wear and residues. ▲ Describes Madjedbebe, Mirarr Country and grinding within deep Aboriginal history. ● Explains the debate by connecting each interpretation to a different evidence base. ◆ Compares independent evidence for function, then asks targeted reliability questions about contamination and layer movement. ★ Identifies which methods added independent support, and preserves the unresolved questions for further inquiry.	■ Explains why the peer-reviewed figure combines several records, and which audience and argument it addresses. ▲ Explains the deep-history research context and the role of Mirarr custodial permission. ● Analyses each interpretation through its evidence: wear and residue, site integrity and dating, or termite and sediment movement. ◆ Identifies grinding as a stronger claim than exact plant or exact age, and explains why. ★ Monitors whether evidence is merely listed, then deliberately revises the list into a comparative judgement.	■ Analyses explicit evidence, the publication's implicit argument, its purpose and its scientific audience. ▲ Uses context to explain the creators' values: empirical testing, custodial permission and careful qualification. ● Exposes why interpretations differ — researchers give different weight to converging records and site-formation evidence. ◆ Weighs a strength, three limitations, and two claims that carry different degrees of certainty. ★ Justifies triangulation, then adapts the conclusion rather than forcing all claims to sound equally certain.

LEARNING CONTINUUM	GRADE 5	YEAR 6	YEAR 7	YEAR 8	YEAR 9	YEAR 10
ORIGIN, FEATURES & PURPOSE	I can name a historical source and a feature I notice.	I can identify different types of historical sources.	I can identify the origin, content features and purpose of historical sources.	I can describe the origin, content features and purpose of historical sources.	I can explain the origin, content features and purpose of historical sources.	I can analyse the origin, explicit and implicit meaning, and purpose of historical sources.
HISTORICAL CONTEXT	I can say where or when a source comes from.	I can identify the historical context of a source.	I can outline the historical context of a source.	I can describe the historical context of sources.	I can explain historical context, including important events that influenced a source's creation.	I can explain historical context to identify the motivations, values and attitudes of a source's creator.
HISTORICAL INTERPRETATIONS	I can notice that people may explain the past differently.	I can recognise that historians may interpret historical events and developments differently.	I can describe different historical interpretations of the past.	I can explain different historical interpretations and contested debates about the past.	I can analyse historical interpretations and contested debates and identify the evidence used to support them.	I can evaluate historical interpretations and contested debates, including why or how interpretations differ.
ACCURACY, USEFULNESS & RELIABILITY	I can say one thing a source can and cannot tell us.	I can recognise that some historical sources are more useful than others.	I can draw conclusions about the usefulness of sources.	I can compare and contrast historical sources and ask questions about their accuracy, usefulness and reliability.	I can compare and contrast historical sources and identify their accuracy, usefulness and reliability.	I can compare and contrast sources and analyse their accuracy, usefulness and reliability.
METACOGNITION	I can follow a source-analysis checklist one step at a time.	I can complete the task by breaking it into smaller parts.	I can review and edit my work.	I can reflect on my learning experience and processes to inform future progress.	I can use, monitor and evaluate a variety of learning strategies.	I can justify and adapt the source-analysis strategies that produced my strongest judgement.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), "65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia," *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.